

ACADMATE

SPRINT 1 | CORE ADVISEMENT PLATFORM

AGENDA

Sprint 1 review

A focused review of the product, sprint evidence and next increment.

01 Product & team

Scope, personas, ownership

02 Solution design

Architecture, data model, technology

03 Sprint execution

Backlogs, data, tests and status

04 Outcomes & next steps

Metrics, demo, retrospective, Sprint 2

PEOPLE + COLLABORATION

Team ownership

Tafadzwa
Product Owner • AI/ML



Pranali
Scrum Master • Backend



Lyno
Frontend • LLM



Liangxian Li
Backend



Datta
Database • Backend



Saisai
UI/UX • Documentation



Chintu
Frontend • QA



Improvements from Professor Feedback

The professor advised our team to give individual introductions while presenting so that each member is properly introduced in the presentation keep the presentation seem like a one team.

Writing actionable retrospective points clearly and presenting how we conducted the retrospective.

Project Description

A full-stack advising platform that turns academic data into actionable support



Product, personas & MVP

Academic records become a usable foundation for student and advisor decisions.



STUDENT (Lucas Vance, 20y)

"I want to stay on track and reach my goals."

- *Fragmented academic information*
- *Late awareness of declining performance*
- *Generic rather than personalized support*



ADVISOR (Elena Sterling Academic Advisor, 38y)

"I want to support my students before they fall behind."

- *Large advising caseloads*
- *Manual review of multiple data sources*
- *Limited time for proactive outreach*



ADMINISTRATOR

(Leo Mercer Administrator, 45)

"I want a safe, reliable system that improves student outcomes."

- *Data privacy and access control*
- *Consistency of advising information*
- *Responsible use of AI*

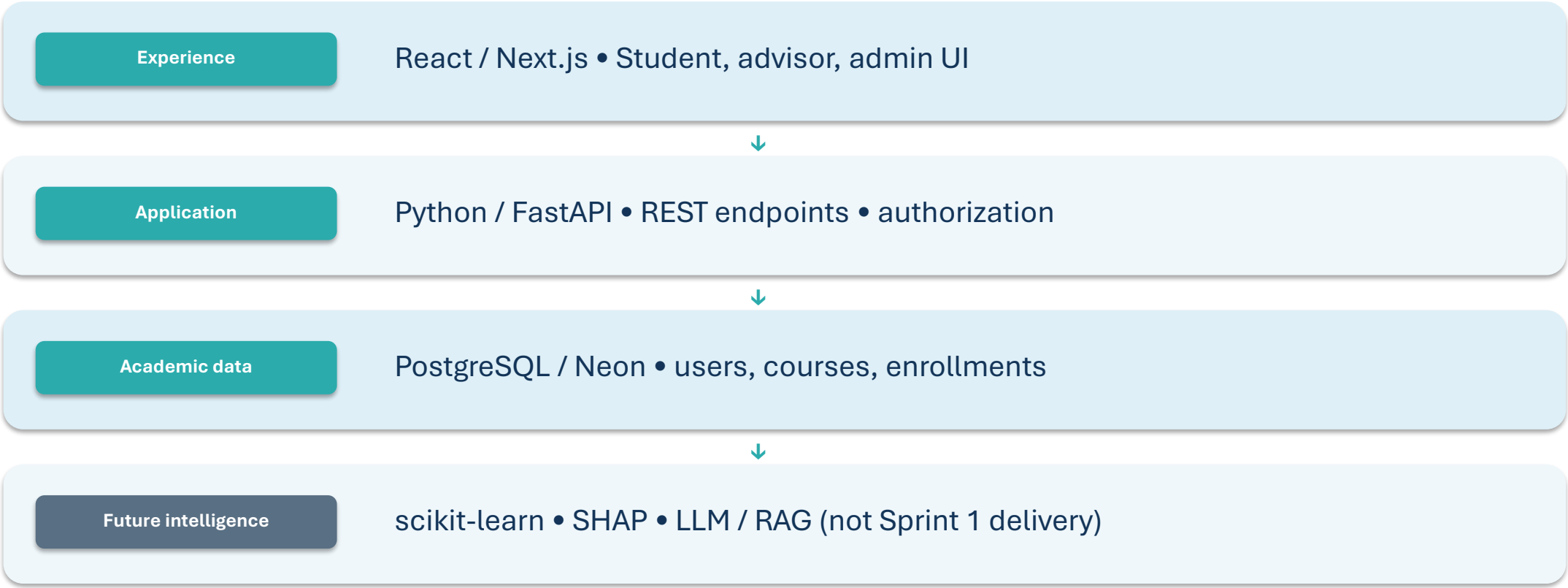
Sprint 1 usable slice

Sign in / roles → **Academic records** → **Admin profile** → **Student Profile**

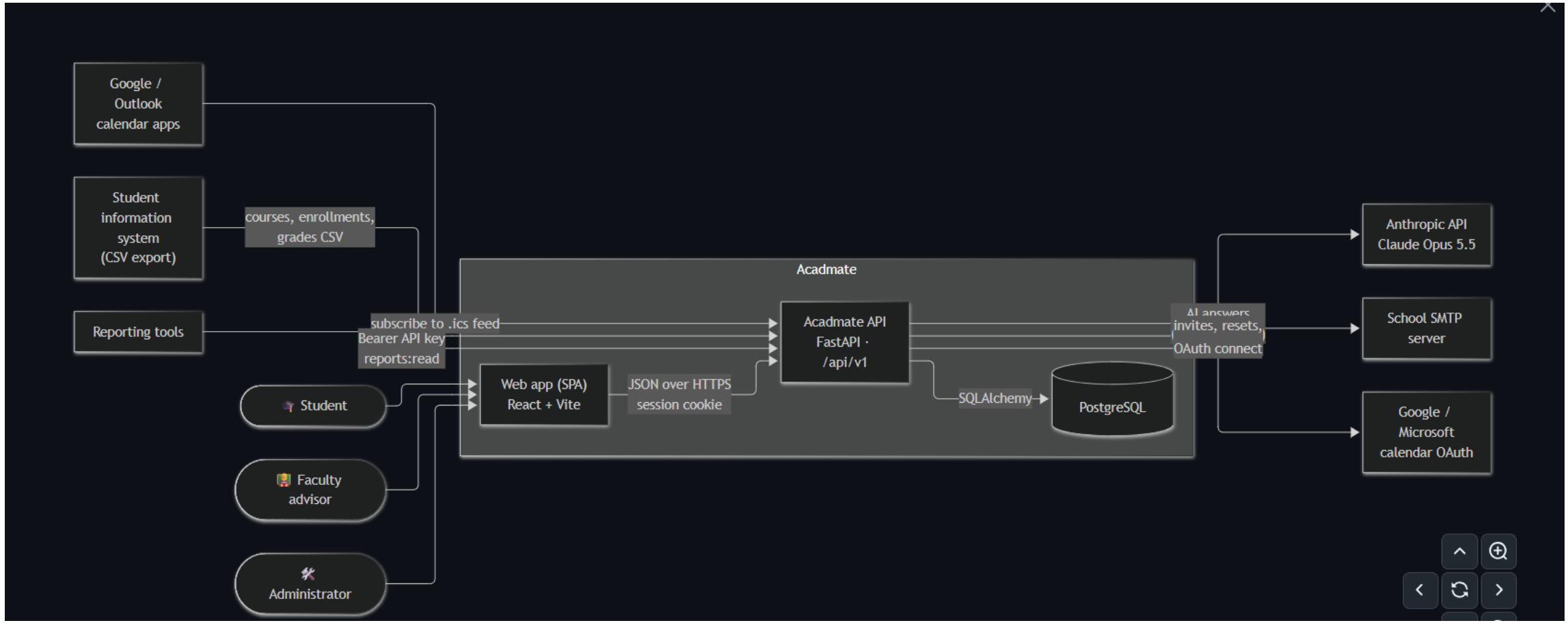
Later increments: Advisor Profile , predictive risk, SHAP explanations and RAG-supported advising.

Architecture & technology

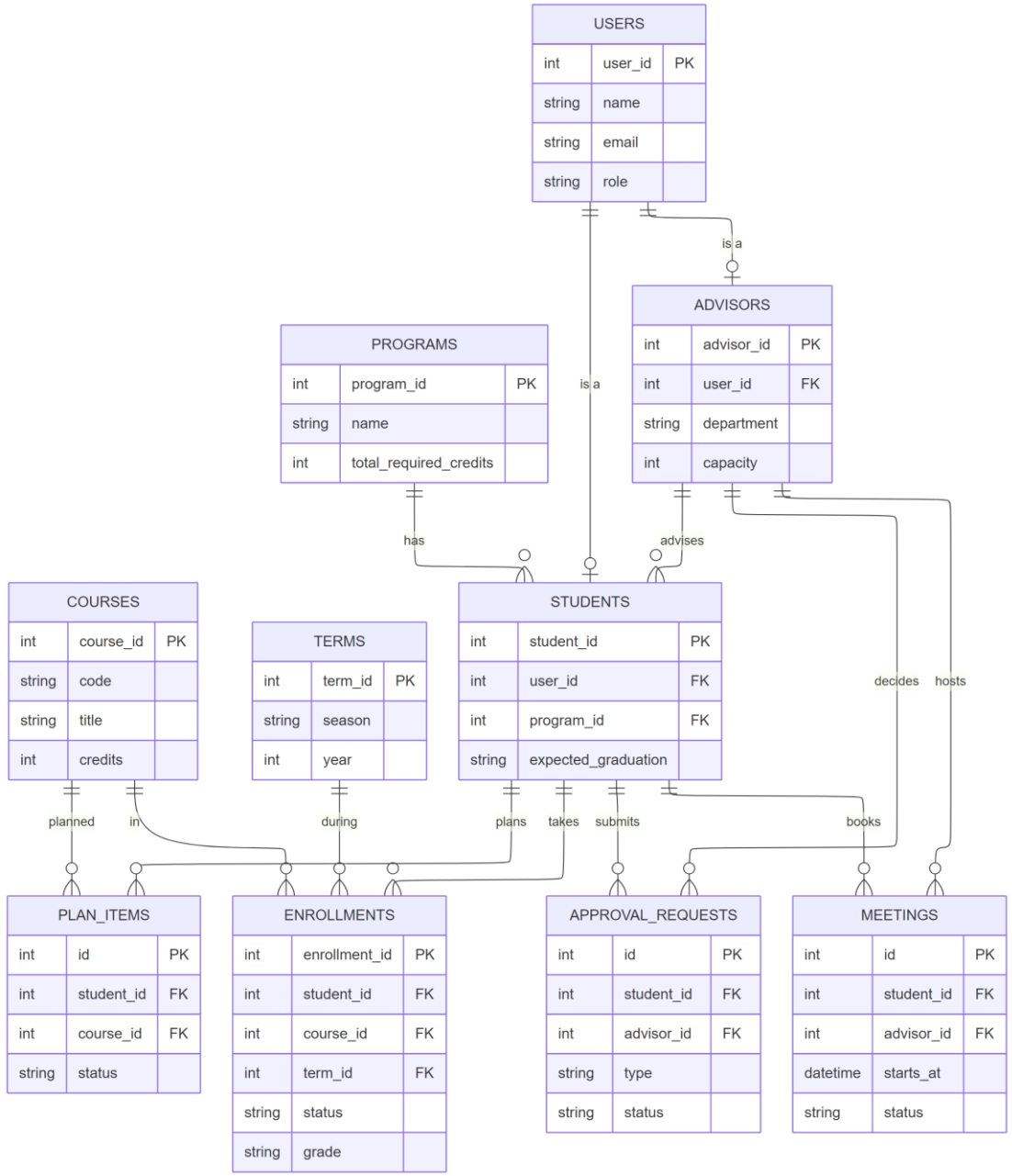
Core Sprint 1 services are separated from future intelligence capabilities.



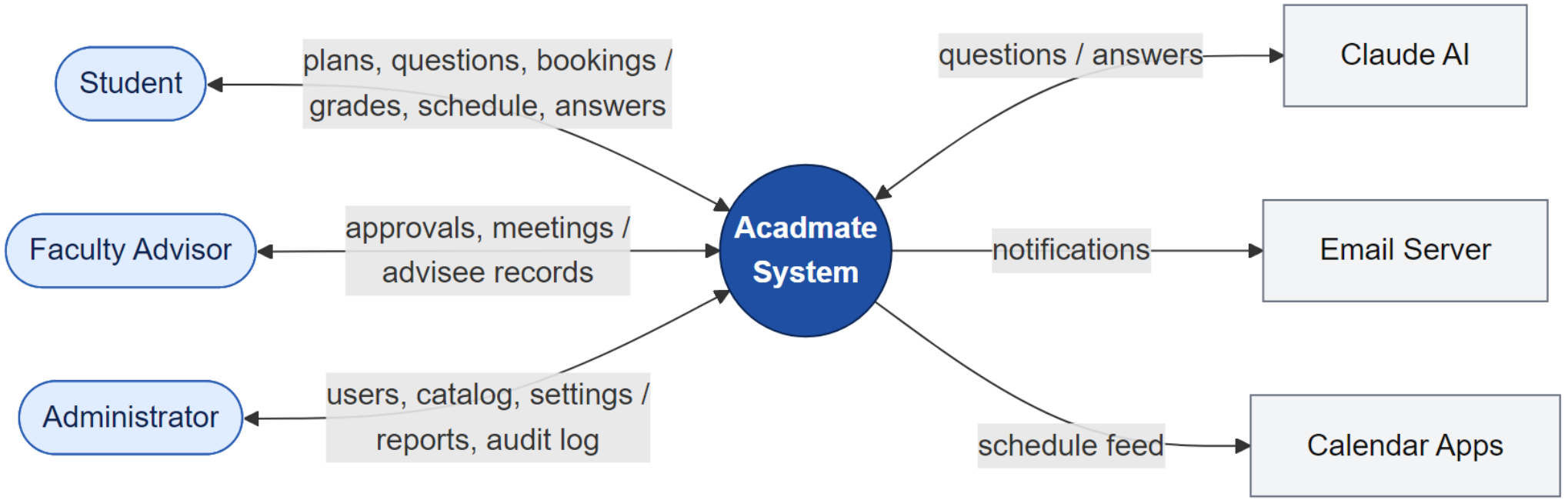
Architecture & technology



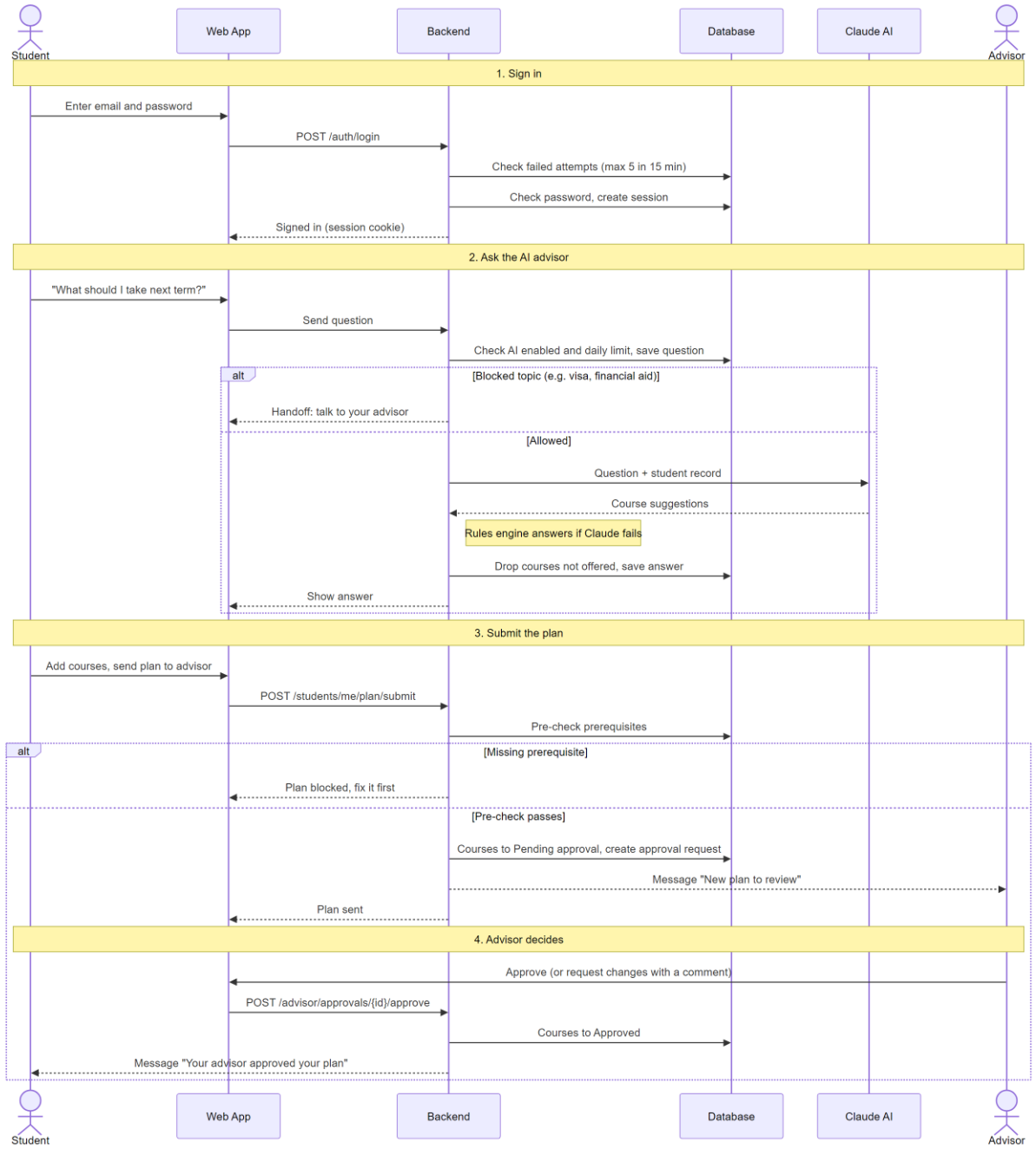
ER Diagram


USERS
int user_id PK
string name
string email
string role
is a
ADVISORS
int advisor_id PK
int user_id FK
string department
int capacity
PROGRAMS
int program_id PK
string name
int total_required_credits
is a
has
advises
COURSES
int course_id PK
string code
string title
int credits
TERMS
int term_id PK
string season
int year
STUDENTS
int student_id PK
int user_id FK
int program_id FK
string expected_graduation
decides
hosts
planned
in
during
plans
takes
submits
books
PLAN_ITEMS
int id PK
int student_id FK
int course_id FK
string status
ENROLLMENTS
int enrollment_id PK
int student_id FK
int course_id FK
int term_id FK
string status
string grade
APPROVAL_REQUESTS
int id PK
int student_id FK
int advisor_id FK
string type
string status
MEETINGS
int id PK
int student_id FK
int advisor_id FK
datetime starts_at
string status

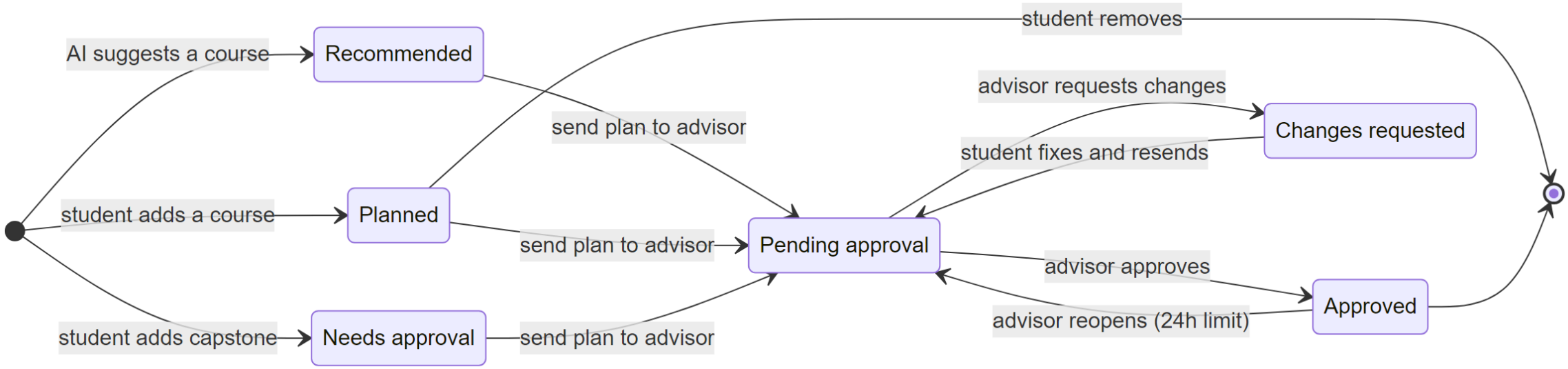
Context Diagram



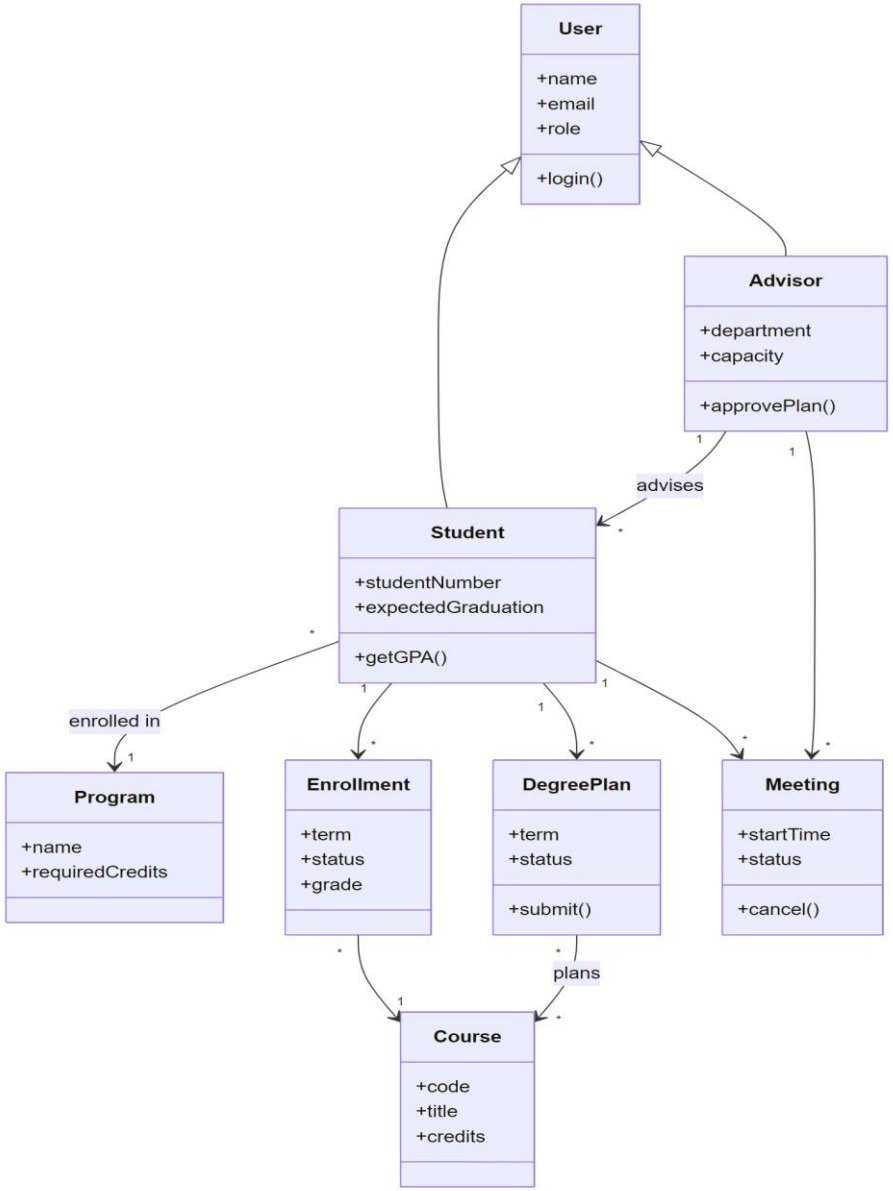
Sequence Diagram



State Diagrams

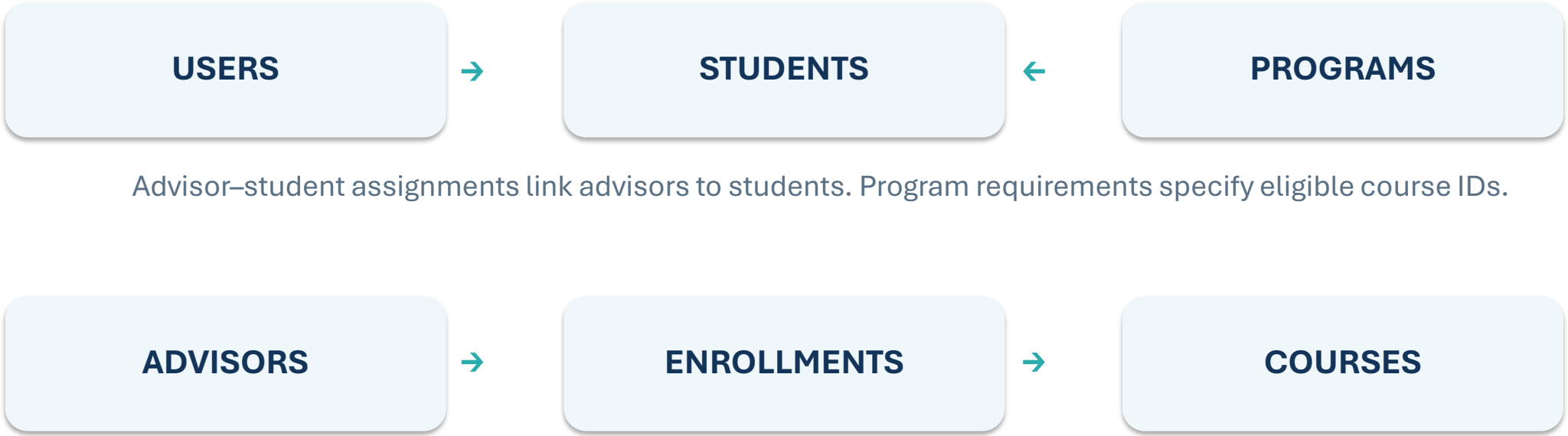


Class Diagrams



Academic data model

The synthetic CSVs mirror the backend's core entities and relationships.



Advisor–student assignments link advisors to students. Program requirements specify eligible course IDs.

Contract fields: enrollment.status = in_progress • program_requirements.eligible_course_ids

Product backlog

The complete product backlog spans the full MVP; Sprint 1 selects the foundational work.

ID	Product Backlog Item	Description	Priority
PB-01	User Authentication and Access Management	Implement secure login and logout, user registration where applicable, password management, and role-based access for students, advisors, and administrators. Ensure users can access only the features and data permitted for their roles.	High
PB-02	Data Import and Management	Enable importing student data from CSV or Excel files, including student profiles, grades, attendance, course enrollment, assignment submissions, and LMS engagement. Validate imported data, handle missing or duplicate records, and provide import status and error reports.	High
PB-03	Student Profile and Dashboard Management	Develop a centralized dashboard displaying academic performance, course enrollment, attendance, assignment submissions, predicted academic risk levels, and personalized recommendations. Allow students to monitor their progress and access academic support resources.	High
PB-04	Admin Profile and Dashboard Management	Develop an administrative dashboard to manage student and advisor accounts, monitor overall academic performance, view risk statistics, manage system settings, and generate reports on student engagement and advising activities.	High
PB-05	Advisor Profile and Dashboard Management	Develop advisor profiles and a centralized dashboard to view student information, academic history, course enrollment, attendance, academic progress, risk levels, and pending interventions.	High
PB-06	AI-Based Student Risk Prediction and Explainability	Develop an AI model that analyzes grades, attendance, assignment submissions, and LMS engagement to identify at-risk students and explain the key factors contributing to predicted risk levels.	High
PB-07	Personalized Recommendations and Academic Interventions	Generate personalized academic recommendations, including tutoring, study plans, and time-management resources. Enable advisors to assign interventions, track progress, schedule follow-ups, and monitor student improvement.	High
PB-08	Conversational AI and RAG Integration	Implement a conversational AI assistant using Retrieval-Augmented Generation (RAG) to answer student and advisor questions using approved university policies, academic documents, course catalogs, and advising resources.	High
PB-09	System Integration and Notifications	Integrate the application with student information systems and learning management systems. Implement automated alerts for declining grades, repeated absences, and missed assignments.	High
PB-10	Testing, Security, and Performance Optimization	Conduct functional, integration, usability, and user acceptance testing. Implement security controls, protect sensitive student data, and conduct performance testing to ensure reliability.	High
PB-11	Deployment, Release, and User Feedback	Deploy the application to the target environment, configure production settings, prepare release documentation and user guidance, and collect student and advisor feedback to identify improvements for future releases.	High

Sprint 1 backlog

Data & import	Schema-aligned CSVs; valid keys, statuses and repeat scenarios
Identity & access	Authenticated roles; student/advisor records protected
Student academic profile	Course history and GPA/credits exposed correctly
Admin workflow	Assigned-student roster and dashboard API accessible
AI readiness & review	Risk target/doc sources defined; end-to-end review prepared

SPRINT 1 DELIVERABLE

Synthetic academic dataset

Fixed CSV artifacts prepared for repeatable integration testing.

100
students

800
enrollments

30
courses

4
advisors

Coverage

105 users • 100 assignments • 1 program • 3 requirement groups

Integration handoff

PR #6 reviewed

Canonical fields: in_progress • eligible_course_ids | Existing seed records preserved.

Sprint 1 User Story and Acceptance Criterial

Acadmate · User stories summary

Epic	Stories	Story Points	High	Medium	Low
E1 · Authentication & Access	2	13	2	0	0
E2 · Student Experience	7	55	5	2	0
E3 · Administration	4	42	4	0	0
Total	13	110	11	2	0

Sprint 1 Test Cases

Separate verified dataset work from integration work requiring final team confirmation.

Acadmate · Test case summary

Story	Feature	Test cases	Functional	API	Security	Regression	High	Medium	Low
US-01	Sign-in, sessions & navigation	2	1	0	1	0	2	0	0
US-02	Account setup & recovery	1	0	1	0	0	1	0	0
US-03	Dashboard, courses & grades	1	1	0	0	0	1	0	0
US-04	Degree plan & approval	1	0	1	0	0	1	0	0
US-05	AI advisor	1	0	0	1	0	1	0	0
US-06	Schedule & assignments	1	1	0	0	0	1	0	0
US-07	Messages & notices	1	0	0	1	0	1	0	0
US-08	Book a meeting	2	1	1	0	0	2	0	0
US-09	Student settings & privacy	1	0	0	1	0	1	0	0
US-10	Overview & audit log	1	1	0	0	0	1	0	0
US-11	Users & advisors	1	0	1	0	0	1	0	0
US-12	Catalog & data integrations	1	1	0	0	0	1	0	0
US-13	AI & institution settings	1	0	1	0	0	0	1	0
Total		**15**	**6**	**5**	**4**	**0**	**14**	**1**	**0**

Sprint 1 metrics

Use the team's recorded story points; do not substitute issue counts or estimates.

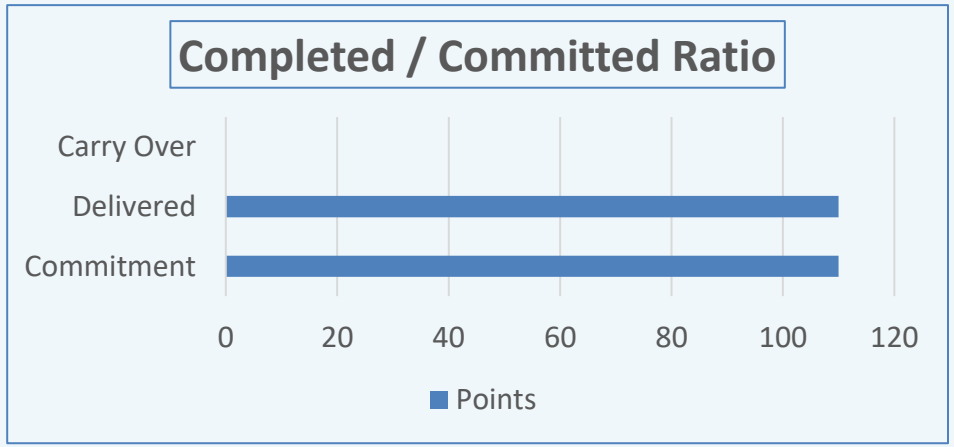
Velocity / completion

Estimated Points : 110

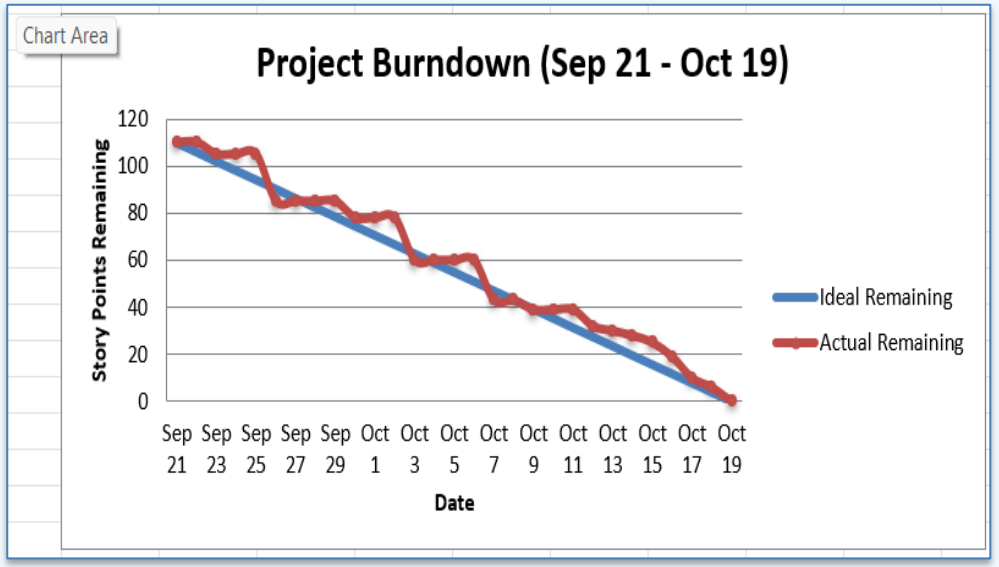
Total Points Delivered : 110

Total Point Carried Over : 0

Team Velocity: 110



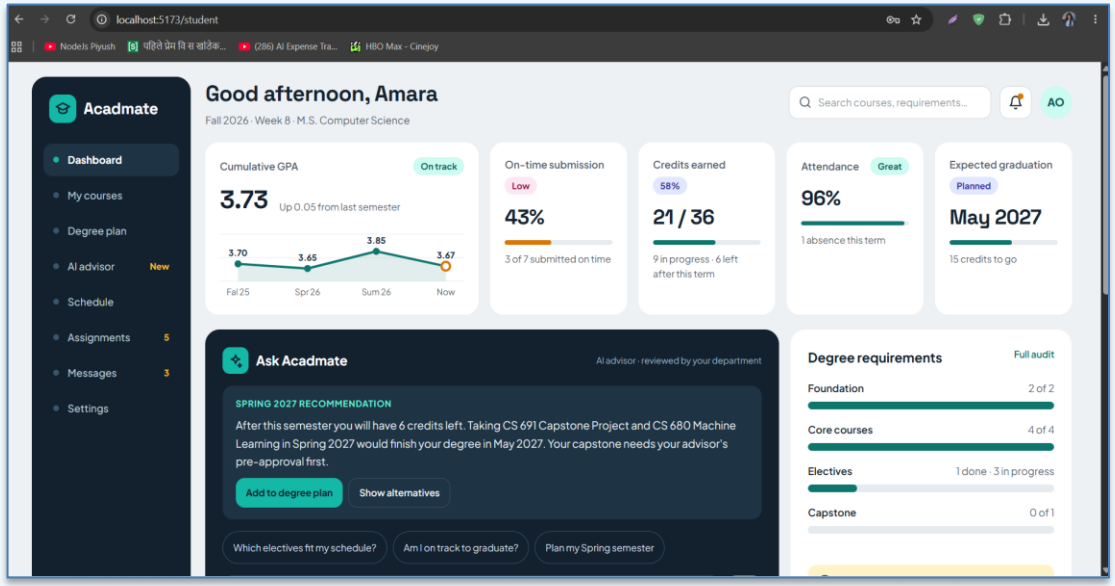
Sprint 1 burndown



Sprint 1 product demonstration

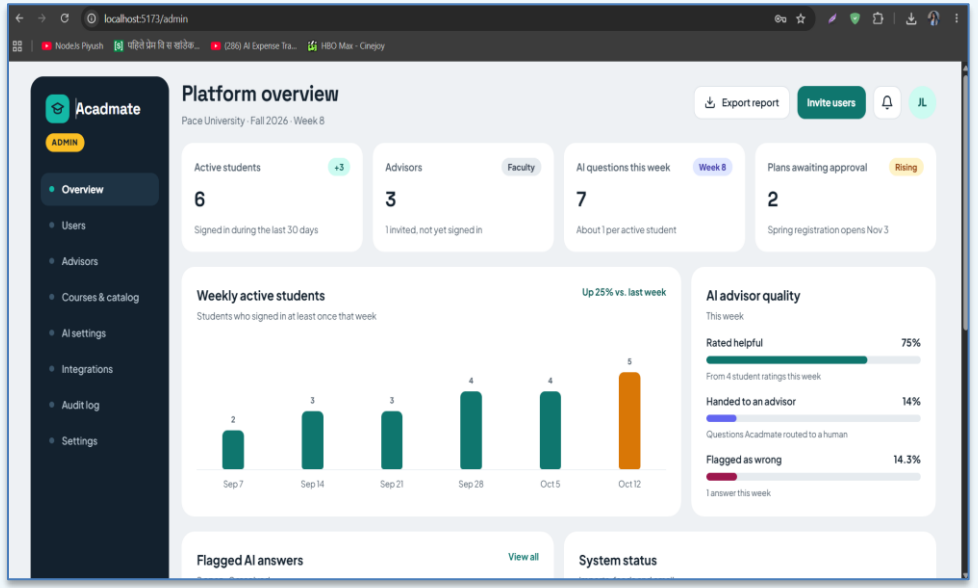
Capture only real screens from the integrated application, not mockups.

STUDENT UI



Academic profile / course history

ADMIN UI



Overview of Advisor-student performance

Repository: github.com/htmw/F2026-Team3

Team working agreement & retrospective

Shared responsibilities, clear acceptance criteria and earlier integration.

Working agreement

- Collaborated effectively on synthetic dataset ownership and schema alignment.
- Conducted peer reviews across backend and data development.
- Completed assigned tasks successfully and kept the GitHub repository organized and clutter-free.
- Ensured smooth integration of work and completed all sprint tasks within the desired timeline.

What worked

- Each team member completed their assigned tasks successfully and on time.
- We maintained clear ownership of the synthetic dataset and aligned the schemas across backend and data development.
- We kept the GitHub repository organized and clutter-free, with peer reviews to ensure code quality and smooth integration.
- Through effective teamwork and communication, we completed all sprint tasks within the desired timeline.

Improve

- The team collaborated on implementation and completed the assigned tasks, but some work continued until the last day of the sprint.
- We had limited time for the final demo preparation and rechecking APIs and routes.
- A few meetings were missed, which affected coordination and progress tracking.

Next actions

1. Complete tasks earlier: Set clear deadlines for each team member and aim to finish implementation before the last day, leaving enough time for API and route testing, bug fixing, and demo preparation.
2. Improve coordination and demo readiness: Attend scheduled meetings consistently, communicate blockers early, and conduct a final team review to ensure all features are integrated, tested, and ready for demonstration.

NEXT SPRINT

Sprint 2 | Planned increment

Carry over uncompleted Sprint 1 work before expanding intelligence features.

01 Advisor Profile Work

Close outstanding auth, advisor UI and integration work

02 Degree progress

Compute requirement-group and credit completion

03 Academic risk

Prepare training dataset, baseline model and thresholds

04 AI foundations

SHAP explanations and approved knowledge ingestion

Thank you

AcadMate: AI-Powered Personalized Academic Advisement System

Questions & discussion

Student data → meaningful academic support